Infinite Repeating Decimals Are Transcendental and Irrational

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Abstract

Based on the transcendence criterion proved by Qi Jin, it can be shown that all infinite repeating decimals are transcendental numbers, and hence irrational. Consequently, infinite repeating decimals are not rational numbers. Treating repeating decimals as rational numbers is a logical error. The infinite repeating decimal 0.(9) and the number 1 are two distinct numbers, even though the difference between them tends to zero. Finally, it is proved that all infinite decimals are transcendental numbers.

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1 Definitions:

- A rational number is a fraction whose numerator and denominator each have finitely many prime factors, and the exponent of each prime factor is also finite.
- An irrational number is a number that is not rational, such as $\sqrt{2}$ and π.

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- A transcendental number is a number that does not satisfy any nonzero polynomial equation with integer coefficients, that is, a non-algebraic number, such as π.
- All rational numbers are algebraic numbers, so transcendental numbers are not rational numbers.
- All transcendental numbers are irrational numbers.

The relationship among these three types of numbers is shown in the figure below.

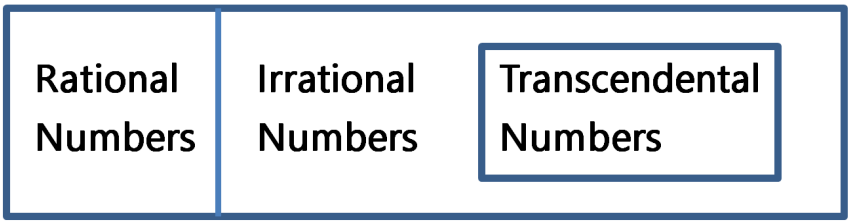


Figure 1: Relationship

2 Equality and Approximation

Consider the following two numbers:

$$\alpha = 0.\dot{9} = 0.9999\ldots = \sum_{n=1}^{\infty} \frac{9}{10^n} \approx 1,$$
$$\beta = \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots \approx 1.$$

Are α and β equal? Are they the same number? Logical analysis shows that α and β are approximately equal but not equal; they are two distinct numbers, and both are transcendental and irrational, not rational. The proof is as follows (where N denotes the largest natural number):

Expressing α with the common denominator 10^N and summing, we obtain:

$$\alpha = \sum_{n=1}^{\infty} \frac{9}{10^n} = 9 \sum_{n=1}^{N} \frac{1}{10^n} = 9\, \frac{10^{N-1} + 10^{N-2} + \cdots + 100 + 10 + 1}{10^N}.$$

The numerator is odd, its units digit is 1, and it has no factors of 2 or 5. The denominator has only the prime factors 2 and 5, so the factor 2^N in the denominator cannot be canceled. According to the transcendence criterion proved by Qi Jin[1],

when the exponent of some prime factor in the numerator or denominator of an irreducible fraction tends to infinity, the fraction is a transcendental number. Here N tends to infinity, so α is a transcendental and irrational number, not a rational number. Hence α and 1 are two distinct numbers, although their distance tends to 0.

Applying the same method to β, and summing with the common denominator 2^N, we obtain:

$$\begin{aligned}\beta &= \sum_{n=1}^{\infty} \frac{1}{2^n} \\ &= \sum_{n=1}^{N} \frac{1}{2^n} \\ &= \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{N-1}} + \frac{1}{2^N} \\ &= \frac{2^{N-1} + 2^{N-2} + \cdots + 4 + 2 + 1}{2^N}.\end{aligned}$$

The numerator of β is odd, and the factor 2^N in the denominator cannot be canceled. According to the transcendence criterion proved by Qi Jin[1], β is a transcendental and irrational number, not a rational number.

For two irreducible fractions, if any prime factor in the numerator or denominator differs, the two fractions cannot be equal. The denominator of α contains the factor 5, whereas the denominator of β does not; therefore α and β are two distinct numbers, two distinct transcendental numbers, and also two distinct irrational numbers. Neither is a rational number, although both of their values tend to 1.

3 Logical Error

Why are repeating decimals, such as $0.\dot{9}$, treated as rational numbers? Because in arithmetic calculations, approximation is mistaken for equality, and infinitesimal quantities can be neglected. However, when studying rational numbers, irrational numbers, and transcendental numbers, infinitesimal quantities cannot be ignored.

Consider the two numbers $0.\dot{3}\dot{5}$ and $\frac{35}{99}$. The repeating decimal $0.\dot{3}\dot{5}$ is actually the sum of an infinite geometric series with first term 0.35 and common ratio $\frac{1}{100}$, which gives:

$$0.\dot{3}\dot{5} = 0.35\,\frac{1 - \left(\frac{1}{100}\right)^N}{1 - \frac{1}{100}}.$$

Since N tends to infinity, $\left(\frac{1}{100}\right)^N \to 0$. After neglecting $\left(\frac{1}{100}\right)^N$, we obtain:

$$0.35\,\frac{1-\left(\frac{1}{100}\right)^N}{1-\frac{1}{100}} = 0.35\,\frac{1}{\frac{99}{100}} = \frac{35}{99}.$$

Thus $0.\dot{3}\dot{5}$ and $\frac{35}{99}$ are approximately equal, not equal. When determining whether a number is rational, this approximate result is used directly, treating approximation as equality, which leads to the fallacy that repeating decimals are rational numbers.

Regardless of the theoretical framework, repeating decimals are transcendental numbers, irrational numbers, and not rational numbers. This is a logical matter, not a matter of choosing a number system.

4 $\infty \neq \infty - 1$

There is a common misunderstanding often used to argue that 0.(9) is an algebraic number, namely the claim that 0.(9) satisfies the equation

$$10x - x = 9.$$

The reasoning is: $10 \times 0.999\cdots = 9.999\cdots$, so $10x - x = 9$.

However, 0.(9) has infinitely many decimal places. The logical error here lies in assuming that the decimal part of $10x$ is equal to the decimal part of x; in other words, it assumes that $\infty = \infty - 1$. After multiplying x by 10, the number of decimal places becomes one fewer than that of x, so the two are not equal. In fact, 0.(9) satisfies the equation

$$10x - x = 9 - 9 \times 10^{-\infty} \neq 9.$$

If one assumes $\infty = \infty - 1$, it leads to the following error:

Let

$$S = 1 + 2 + 4 + 8 + \cdots + 2^{\infty}.$$

Then one obtains:

$$\begin{aligned} & S = 1 + (2 + 4 + 8 + \cdots + 2^{\infty}) \\ \Rightarrow \quad & S = 1 + 2\big(1 + 2 + 4 + \cdots + 2^{\infty-1}\big) \\ \Rightarrow \quad & S = 1 + 2S \\ \Rightarrow \quad & S = -1. \end{aligned}$$

Therefore, only by assuming $\infty \neq \infty - 1$ can such an absurd result be avoided.

5 Infinite Decimals Are Transcendental

For any infinite decimal $\theta < 1$, expressed as $\theta = 0.C_1C_2C_3\cdots C_k$, where $k \to \infty$ and $C_k \neq 0$ (with $C_i \in \{0,1,2,3,4,5,6,7,8,9\}$ representing the i-th digit after the decimal point), it can be written as a fraction as follows:

$$\theta = \frac{C_1C_2C_3\cdots C_k}{10^k} = \frac{C_1C_2C_3\cdots C_k}{2^k \cdot 5^k}, \quad C_k \neq 0.$$

Since $C_k \in \{1,2,3,4,5,6,7,8,9\}$, the numerator of θ cannot simultaneously contain both 2 and 5 as factors. Consequently, at least one of the terms 2^k or 5^k in the denominator remains unaffected by simplification. According to the transcendental number criterion proved by Qi Jin[1], it follows that θ must be a transcendental number.

In other words, all infinite decimals, whether repeating or non-repeating, are transcendental numbers.

If there are N natural numbers, and infinite decimals also have N digits, then there are only 10^N decimals in the interval $(0,1)$. However, Qi Jin has proved that there are $N^{N^{N^{\cdot^{\cdot^{\cdot}}}}}$ (with infinitely many copies of N) real numbers between 0 and 1[2]. Therefore, many real numbers cannot be represented exactly by decimals, but can only be approximated.

Algebraic irrational numbers, such as $\sqrt{2}$, cannot be represented exactly by decimals, but can only be approximated by infinite decimals.

References

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[2] Jin, Q. (2026). *Proof of the Continuum Hypothesis*, preprint. Zenodo: `https://doi.org/10.5281/zenodo.20651018`.